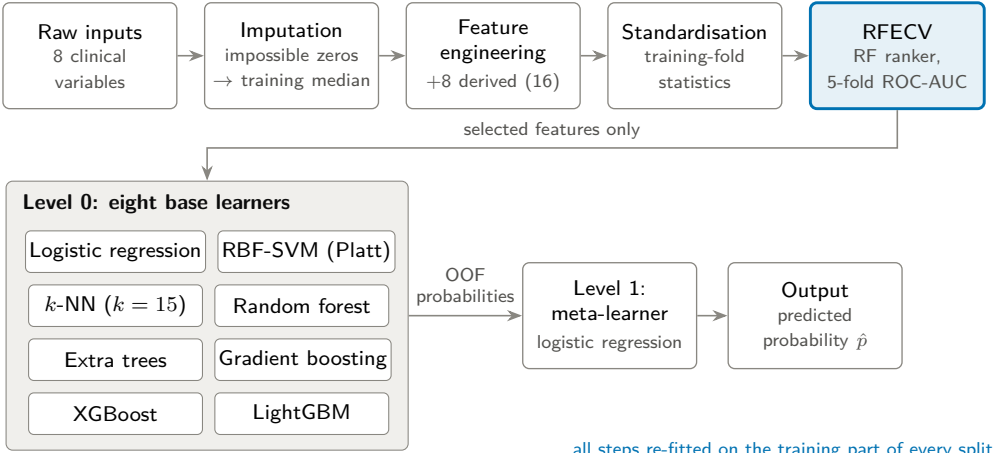

Raw inputs
8 clinical variables
Imputation
impossible zeros → training median
Feature engineering
+8 derived (16)
Standardisation
training-fold statistics
RFECV
RF ranker, 5-fold ROC-AUC
selected features only
Level 0: eight base learners
Logistic regression
RBF-SVM (Platt)
k-NN (k = 15)
Random forest
Extra trees
Gradient boosting
XGBoost
LightGBM
OOF probabilities
Level 1: meta-learner
logistic regression
Output
predicted probability p̂
all steps re-fitted on the training part of every split